

**Aim :** Demonstrate working of a blockchain. Create a blockchain in Python.

**Code :**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> import hashlib import time  class Block:     def __init__(self, index, timestamp, data, prior_hash="0"):         self.index = index         self.timestamp = timestamp         self.data = data         self.prior_hash = prior_hash         self.hash = ""      def createhash(self):         blockString = f"{self.index}{self.timestamp}{self.data}{self.prior_ha sh}".encode()         return hashlib.sha256(blockString).hexdigest()      def display(self):         print(f"Index    : {self.index}")         print(f"Timestamp : {self.timestamp}")         print(f"Data      : {self.data}")         print(f"Prior Hash : {self.prior_hash}")         print(f"Hash       : {self.hash}")         print("-" * 60)  class Blockchain:     def __init__(self):         self.chain = [self.genesis_block()]      def genesis_block(self):         block = Block(0,time.strftime("%H:%M:%S", time.localtime()),"Genesis Block","0")         block.hash = block.createhash()         return block      def get_lastblock(self): </pre> | <pre>         return self.chain[-1]      def append_block(self, block):         block.prior_hash = self.get_lastblock().hash         block.hash = block.createhash()         self.chain.append(block)      def checkChain(self):         for i in range(1, len(self.chain)):             curr_block = self.chain[i]             prev_block = self.chain[i - 1]             if curr_block.hash != curr_block.createhash():                 return False             if curr_block.prior_hash != prev_block.hash:                 return False         return True      def show_chain(self):         for block in self.chain:             block.display()  Block_Chain = Blockchain() Block1 = Block(1, time.strftime("%H:%M:%S", time.localtime()), "ABC") Block2 = Block(2, time.strftime("%H:%M:%S", time.localtime()), "LMN") Block3 = Block(3, time.strftime("%H:%M:%S", time.localtime()), "XYZ") Block_Chain.append_block(Block1) Block_Chain.append_block(Block2) Block_Chain.append_block(Block3) print("Blockchain Valid:", Block_Chain.checkChain()) print("\nBlockchain:") Block_Chain.show_chain() print("\nChanging Block 2 Data") Block_Chain.chain[2].data = "HACKED" print("Blockchain Valid:", Block_Chain.checkChain()) </pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Output :**

```

Is chain valid? True
Block index 1
Block timestamp 14:28:14
Block data
Prior hash 0
Hash 1660bc6aca502999e71f06836ed7933b7516387099ce110527c439a7970e9c3d

```

**Aim :** Write a smart contract to demonstrate basic data types in solidity.

**Code :**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>// SPDX-License-Identifier: MIT pragma solidity &gt;=0.8.34;  contract Employee {     struct emp {         uint8 emp_id;         string emp_name;         uint256 emp_salary;         string emp_dept;     }      mapping(uint8 =&gt; emp) public idToEmployee;     emp[] public employees;      function addEmployee(uint8 id, string memory nm,         uint256 sal, string memory dep) public {         emp memory a = emp({emp_id: id, emp_name: nm,         emp_salary: sal, emp_dept: dep});         idToEmployee[id] = a;         employees.push(a);     } }</pre> | <pre>function getEmployee(uint8 id) public view returns (emp memory) {     return idToEmployee[id]; }  function addBonus(uint8 id) public {     if (idToEmployee[id].emp_salary &lt; 10000) {         idToEmployee[id].emp_salary += 5000;          for (uint i = 0; i &lt; employees.length; i++) {             if (employees[i].emp_id == id) {                 employees[i].emp_salary =                 idToEmployee[id].emp_salary;                 break;             }         }     } }  function totalEmployees() public view returns (uint) {     return employees.length; } }</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Output :**

● **addEmployee** uint256, string, uint256, string

2

Akash

8000

cs

Calldata [🔗](#) Parameters [🔗](#)

Value 0 wei ▼

Gas limit auto 0

Transact

● **addBonus** uint256

1

Calldata [🔗](#) Parameters [🔗](#)

● **getEmployee** uint256

1

Calldata [🔗](#) Parameters [🔗](#)

0:uint256: 2

1:string: Akash

2:uint256: 13000

3:string: cs

Call

**Aim :** Implement pure and view functions in a Solidity smart contract.

**Code :**

```
// SPDX-License-Identifier: MIT
pragma solidity >=0.8.34;

contract Marks {
    uint public finalMarks;

    function enter_marks(uint Total_Marks, bool NSS) public {
        if (NSS == true) {
            Total_Marks += nssmarks();
        }
        finalMarks = Total_Marks;
    }

    function nssmarks () public pure returns (uint) {
        return 10;
    }

    function get_marks() public view returns (uint) {
        return finalMarks;
    }
}
```

**Output :**

Low level interaction

**enter\_marks** uint16, bool

55

true

Calldata Parameters

Value 0 wei

Gas limit auto 0

Transact

**get\_marks**

0:uint16: 65

Call

Balance 0 ETH

**nssmarks**

0:uint256: 10

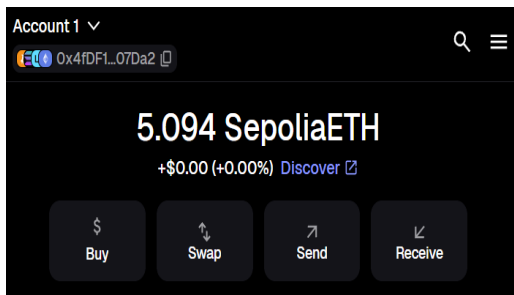
Call

- Aim :** i. Implement payable functions using solidity smart contracts and metamask.  
 ii. Write a solidity smart contract to implement custom errors  
 iii. Demonstrate the usage of Solidity modifiers with a smart contract.

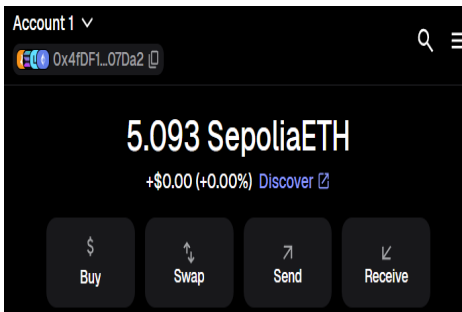
Code :

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>// SPDX-License-Identifier: MIT pragma solidity &gt;=0.8.34;  error InsufficientFunds(string reason, address sender, uint256 amount); error InsufficientDonation(string reason, address sender); contract Donation {     address owner;     constructor() {         owner = msg.sender;     }     modifier onlyOwner() {         require(msg.sender == owner, "You're not the owner");         _;     }     modifier notOwner() {         require(msg.sender != owner, "Owner cannot donate");         _;     } }</pre> | <pre>function donate() public payable notOwner {     if (msg.sender.balance &lt;= 100000 gwei) {         revert InsufficientFunds("Donor balance must be above 100K Gwei", msg.sender, msg.sender.balance);     }     if (msg.value &lt; 5 gwei) {         revert InsufficientDonation("Minimum donation is 5 Gwei", msg.sender);     } } function withdraw() public payable onlyOwner returns (bool) {     uint256 acc_bal = address(this).balance;     if (acc_bal == 0) {         revert InsufficientFunds("No funds available to withdraw", msg.sender, acc_bal);     }     (bool success, ) = owner.call{value: acc_bal}("");     return success; }</pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

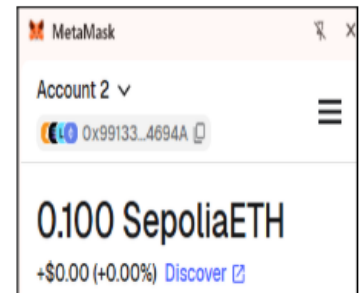
Output: 1) Donation and Withdraw  
Before Contract Deployment



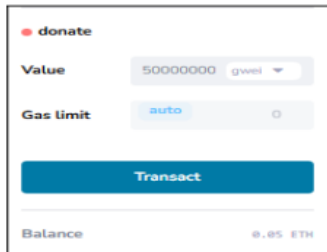
After Contract Deployment



Before Donation



Contract Balance



After Donation



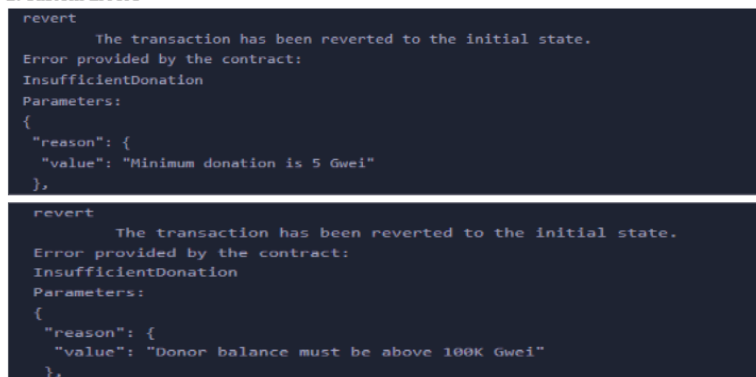
Before Withdraw



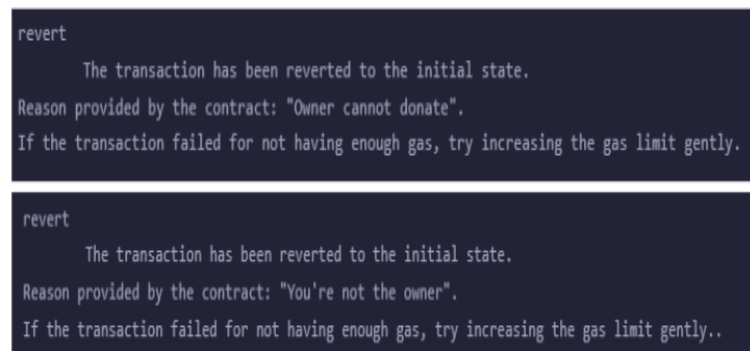
After Withdraw



## 2. Custom Errors



## 3. Modifiers



**Aim :** Write a smart contract to demonstrate the usage of emit-event in solidity.

**Code :**

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.0;
```

```
error InsufficientBalance(string reason , address sender);
```

```
contract P7{
```

```
    event dontaionMorethan10ETH(string MSG , address sender ,uint256 amount);
```

```
    address owner;
```

```
    function donation () public payable {
```

```
        if (msg.value < 5 ether){
```

```
            revert InsufficientBalance("Donation Amount Must Be More Than 5 ethers",owner);
```

```
        }
```

```
        if (msg.value > 10 ether){
```

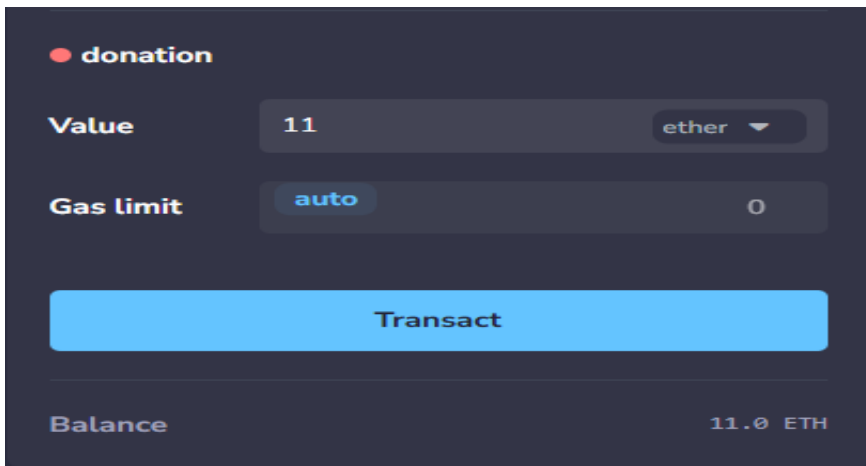
```
            emit dontaionMorethan10ETH("Thank You for Donation", owner , msg.value);
```

```
        }
```

```
    }
```

```
}
```

**Output :**



donation

Value 11 ether

Gas limit auto 0

Transact

Balance 11.0 ETH

```
logs [
    {
        "from": "0x1c91347f2A44538ce624538EBd9Aa907C662b4bD",
        "topic": "0xfc68c6195a27f38561850aaac1728944af402c607b8aead016bdd1f4c5663806",
        "event": "dontaionMorethan10ETH",
        "args": {
            "0": "Thank You for Donation",
            "1": "0x0000000000000000000000000000000000000000000000000000000000000000",
            "2": "11000000000000000000"
        }
    }
]
```

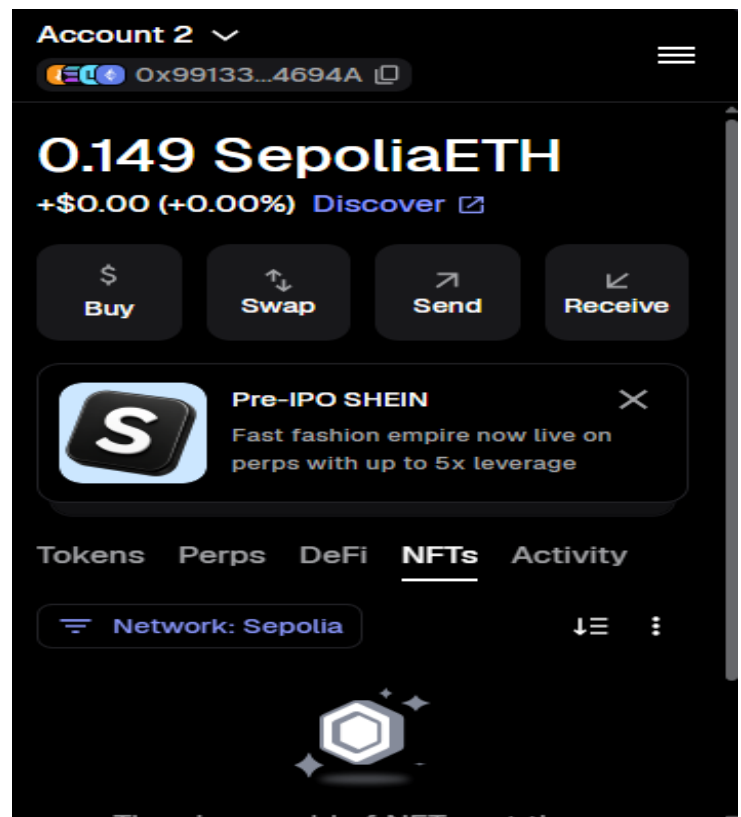
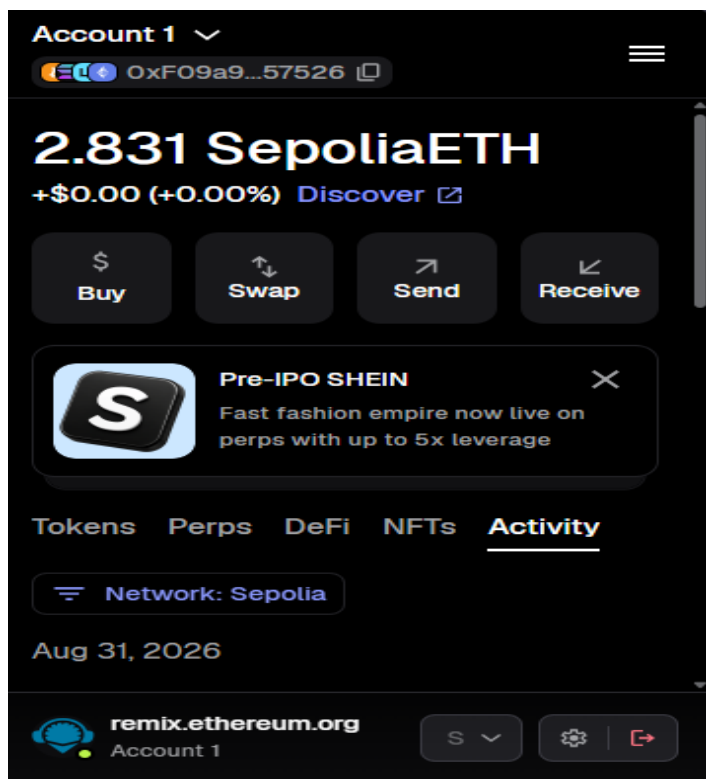
**Aim :** Write a smart contract to mint NFTs. Use IPFS to store tokens.

**Code :**

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.18;
import {ERC721URIStorage, ERC721} from
"@openzeppelin/contracts/token/ERC721/extensions/ERC721URIStorage.sol";
import "@openzeppelin/contracts/access/Ownable.sol";
contract MyNFT is ERC721URIStorage, Ownable {
    uint256 counter;
    constructor() ERC721("TYCS", "TY") Ownable(msg.sender) {}
    function mintNFT(address recipient,string memory tokenURI) public onlyOwner returns
(uint256) {
        counter++;
        uint256 newItemId = counter;
        _mint(recipient, newItemId);
        _setTokenURI(newItemId, tokenURI);
        return newItemId; }}

```

**Output : Before Transaction**



After Transaction(From Account 1 to 2)

Files

Name (A → Z)

103KiB  
MY FILES

3MiB  
LOCAL DATASTORE

+ Import

Name ↑

Pin Status

Size

Screenshot 2026-07-25 092002.png

QmcdwcGcqwQdUrXRLm2SLk1LGNPunBKHKjbRysfmeVvhZ

103 KiB

Share link

MyNFT

0xd44...bbDB0

Sepolia

3min ago

Functions

Select a function to interact with...

Low level interaction

mintNFT address, string

0x9913335dB7Cde3852671463409C03934c3A4f

:qwqQdUrXRLm2SLk1LGNPunBKHKjbRysfmeVvhZ

Calldata

Parameters

Value

0

wei

Account 2

Ox99133...4694A

Buy

Swap

Send

Receive

S

Pre-IPO SHEIN

Fast fashion empire now live on perps with up to 5x leverage

Tokens

Perps

DeFi

NFTs

Activity

Network: Sepolia

**Aim :** Create two smart contracts and import the functions of one contract in the other.

Demonstrate the usage of function modifiers in solidity.

**Code :**

#### Calculator.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.18;
contract Calculator{
    uint256 private result;
    function add(uint256 a,uint256 b)
internal returns (uint256) {
    result = a+b;
    return result; }
    function sub(uint256 a,uint256 b)
internal returns (uint256) {
    result = a-b;
    return result;}
    function mul(uint256 a,uint256 b)
external returns (uint256) {
    result = a*b;
    return result;}
    function div(uint256 a,uint256 b)
external returns (uint256) {
    result = a/b;
    return result; } }
```

#### External.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.18;
import {Calculator} from
'contracts/Calculator_P9.sol';
contract EC {
    uint256 x;
    uint256 y;
    uint256 public r1;
    uint256 public r2;
    function accept_numbers(uint256
num1,uint256 num2) public {
    Calculator cal =
Calculator(0xd9145CCE52D386f2549
17e481eB44e9943F39138);
    x = num1;
    y = num2;
    r1=cal.div(x, y);
    r2=cal.mul(x, y); }
```

#### Internal.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.18;
import {Calculator} from
'contracts/Calculator_P9.sol';
contract Main is Calculator{
    uint256 x;
    uint256 y;
    uint256 public r1;
    uint256 public r2;
    function accept_numbers(uint256
num1,uint256 num2) public {
    x = num1;
    y = num2;
    r1=add(x, y);
    r2=sub(x, y); }}
```

**Output :**

#### External.sol(Div and Mul)

● **accept\_numbers** uint256, uint256

16

8

Calldata  Parameters 

Value 0 wei 

Gas limit auto 0

Transact

● **r1**

0:uint256: 2

Call

● **r2**

0:uint256: 128

Call

#### Internal.sol(Add and Sub)

● **accept\_numbers** uint256, uint256

8

5

Calldata  Parameters 

Value 0 wei 

Gas limit auto 0

Transact

● **r1**

0:uint256: 13

Call

● **r2**

0:uint256: 3

Call

**Aim :** Demonstrate the usage of require and revert statements in solidity.

**Code :**

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.0;
```

```
error InsufficientDonation(string reason, address sender);
```

```
contract P11{
    address owner;
```

```

    function donation () public payable {
        require(msg.sender.balance > 80 ether ,"Account balance Must More Than 80 ethers");
        if (msg.value < 5 ether){
            revert InsufficientDonation("Donation Amount Must Be More Than 5 ethers",owner);
        }
    }
}

```

**Output :**

Account 1 

26.999 ETH

0x5B3...eddC4 

Low level interaction 

● donation

Value  ether 

Gas limit  0

Transact

Account 2 

99.999 ETH

0xAb8...35cb2 

Low level interaction 

● donation

Value  ether 

Gas limit  0

Transact

```
transact to P11.donation errored: Error occurred: Error occurred: revert.
```

```
revert
```

```
The transaction has been reverted to the initial state.
```

```
Reason provided by the contract: "Account balance Must More Than 80 ethers".
```

```
If the transaction failed for not having enough gas, try increasing the gas limit gently..
```

```
Error occurred: revert.
```

```
Error provided by the contract:
```

```
InsufficientDonation
```

```
Parameters:
```

```
{
  "reason": {
    "value": "Donation Amount Must Be More Than 5 ethers"
  },
  "sender": {
    "value": "0x0000000000000000000000000000000000000000"
  }
}
```

**Aim :** A] Install Ganache blockchain test network and deploy solidity smart contract.  
B] Demonstrate setup of myEtherWallet and connect it to Ganache Network.

### Code :

```
// SPDX-License-Identifier: MIT
pragma solidity >=0.8.34;
```

```
contract Marks{
    function enter_marks(uint16 Total_Marks,uint8 nss) public pure returns (uint16) {
        if (nss == 1){
            Total_Marks += 10; // Add 10 marks for NSS
            return Total_Marks;
        }
        return Total_Marks;
    }
}
```

### Output :

#### A] Deploying Contract At External HTTP

The screenshot shows the 'DEPLOY & RUN TRANSACTIONS' panel in the Remix IDE. The environment is set to 'Custom - External Http Provider'. The account 'Account 1' (0x5B3...edcC4) has a balance of 100.000 ETH. The contract 'EC' (p3.sol) is marked as 'Compiled'. The 'Value' is set to 0 and the 'Gas limit' is set to 'auto'. The 'Deploy' button is visible. On the right, the 'External Http Provider' dialog is open, showing the command to run Geth with the --http and --http.corsdomain flags. A warning message states: 'It is not safe to use the --http.corsdomain flag with a wildcard: --http.corsdomain \*'. The 'External HTTP Provider Endpoint' is set to 'http://127.0.0.1:7545'.

The screenshot shows the Ganache CLI interface. The top bar displays various metrics: CURRENT BLOCK 2, GAS PRICE 20000000000, GAS LIMIT 6721975, HARDFORK MERGE, NETWORK ID 5777, RPC SERVER HTTP://127.0.0.1:7545, MINING STATUS AUTOMINING, and WORKSPACE QUICKSTART. The MNEMONIC is 'economy stomach tackle rifle feature genre atom draw when human memory survey'. The HD PATH is 'm44'68'0'8account\_index'. The ADDRESS is '0xd562539EeE9E76792aA3E1143cD12d719fF1Db8b'. The BALANCE is '100.00 ETH'. The TX COUNT is 2 and the INDEX is 0. Below this, a table shows the mined blocks:

| BLOCK | MINED ON            | GAS USED | TRANSACTIONS    |
|-------|---------------------|----------|-----------------|
| 1     | 2026-08-27 08:44:18 | 165685   | 1 TRANSACTION   |
| 0     | 2026-08-27 08:43:39 | 0        | NO TRANSACTIONS |

Ganache

ACCOUNTS

BLOCKS

TRANSACTIONS

CONTRACTS

EVENTS

LOGS

SEARCH FOR BLOCK NUMBERS OR TX HQ

CURRENT BLOCK  
1

GAS PRICE  
20000000000

GAS LIMIT  
6721975

HARDFORK  
MERGE

NETWORK ID  
5777

RPC SERVER  
HTTP://127.0.0.1:7545

MINING STATUS  
AUTOMINING

WORKSPACE  
QUICKSTART

SAVE

SWITCH

TX HASH  
0x817625acfe64a429bda21eea0c3b2bc9b702979569b4c5c9663a84e6b64476fa

CONTRACT CREATION

FROM ADDRESS  
0xd562539EeE9E76792aA3E1143cD12d719fF1Db8b

CREATED CONTRACT ADDRESS  
0x83804784d8D92807028FFAa586d2714899144eee

GAS USED  
165685

VALUE  
0

## B) EtherWallet SetUp

8/27/26, 8:30 AM

about:blank

My Ether Wallet

www.MyEtherWallet.com

YOUR ADDRESS

Your Address:  
 0x1C5529e32963932361b7Eb04F4c6FE5F64292De6

AMOUNT / NOTES

YOUR PRIVATE KEY

Your Private Key:  
 aee2b7f0facc010d2d8959aa592557c57cd2db260208d0223594a53c92133c56

Always look for this icon when sending to this wallet.

Set Up Your Custom Node

Instructions can be found [here](#)

Node Name

TYCS

URL

http://127.0.0.1

Port

7545

☐ HTTP Basic access authentication

☒ ETH
 ☐ ETC
 ☐ Ropsten
 ☐ Kovan
 ☐ Rinkeby
 ☐ Custom
 ☐ Supports EIP-155

Cancel

Save & Use Custom Node

Check TX Status

During times of high volume (like during ICOs) transactions can be pending for hours, if not days. This tool aims to give you the ability to find and "cancel" / replace these TXs. This is not typically something you can do. It should not be relied upon & will only work when the TX Pools are full. Please, read [about this tool here](#).

0x817625acfe64a429bda21eea0c3b2bc9b702979569b4c5c9663a84e6b64476fa

Check TX Status

Transaction Found

Your transaction was successfully mined and is on the blockchain.

- If you see a red ( ! ), a **BAD INSTRUCTION** or **OUT OF GAS** error message, it means that the transaction was not successfully sent. You cannot cancel or replace this transaction. Instead, send a new transaction. If you received an "Out of Gas" error, you should double the gas limit you specified originally.
- If you do not see any errors, your transaction was successfully sent. Your ETH or Tokens are where you sent them. If you cannot see this ETH or Tokens credited in your other wallet / exchange account, and it has been 24+ hours since you sent, please [contact that service](#). Send them the [link](#) to your transaction and ask them, nicely, to look into your situation.

INPUT:

|                                                                                                                                           |                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>{   "config": {     "chainId": 987,     "homesteadBlock": 0,     "eip150Block": 0,     "eip155Block": 0,     "eip158Block": 0,</pre> | <pre>    "terminalTotalDifficulty": 0,     "terminalTotalDifficultyPassed": true   },   "difficulty": "0x400",   "gasLimit": "0x8000000",   "alloc": {} }</pre> |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

```

Administrator: C:\Windows\System32\cmd.exe - geth --datadir E:\MyNetwork\MyBlockchain -networkid 8090
INFO [08-31 09:32:58.467] Commit new sealing work number=1151 sealhash=003d2e..817705 uncles=0 txs=0 gas=0 fees=0 elapsed=6.691ms
INFO [08-31 09:32:58.470] Commit new sealing work number=1151 sealhash=003d2e..817705 uncles=0 txs=0 gas=0 fees=0 elapsed=10.272ms
INFO [08-31 09:32:58.475]   mined potential block number=1150 hash=62082c..517c42
INFO [08-31 09:32:58.773] Successfully sealed new block number=1151 sealhash=003d2e..817705 hash=e6f558..9f9c64 elapsed=311.840ms
INFO [08-31 09:32:58.773]   block reached canonical chain number=1144 hash=dcbfed..0010b0
INFO [08-31 09:32:58.778] Commit new sealing work number=1152 sealhash=65746d..8c4f53 uncles=0 txs=0 gas=0 fees=0 elapsed=5.135ms
INFO [08-31 09:32:58.782] Commit new sealing work number=1152 sealhash=65746d..8c4f53 uncles=0 txs=0 gas=0 fees=0 elapsed=9.787ms
INFO [08-31 09:32:58.786]   mined potential block number=1151 hash=e6f558..9f9c64
INFO [08-31 09:32:59.154] Successfully sealed new block number=1152 sealhash=65746d..8c4f53 hash=2c54f9..4bc304 elapsed=381.664ms
INFO [08-31 09:32:59.154]   block reached canonical chain number=1145 hash=a04543..5ecadb
INFO [08-31 09:32:59.161] Commit new sealing work number=1153 sealhash=0c7809..501874 uncles=0 txs=0 gas=0 fees=0 elapsed=6.819ms
INFO [08-31 09:32:59.164] Commit new sealing work number=1153 sealhash=0c7809..501874 uncles=0 txs=0 gas=0 fees=0 elapsed=10.102ms
INFO [08-31 09:32:59.168]   mined potential block number=1152 hash=2c54f9..4bc304
INFO [08-31 09:32:59.219] Successfully sealed new block number=1153 sealhash=0c7809..501874 hash=52238f..df4608 elapsed=64.887ms
INFO [08-31 09:32:59.219]   block reached canonical chain number=1146 hash=50d885..a2e37e
INFO [08-31 09:32:59.226] Commit new sealing work number=1154 sealhash=3007ac..490bec uncles=0 txs=0 gas=0 fees=0 elapsed=7.258ms
INFO [08-31 09:32:59.230] Commit new sealing work number=1154 sealhash=3007ac..490bec uncles=0 txs=0 gas=0 fees=0 elapsed=11.097ms
INFO [08-31 09:32:59.233]   mined potential block number=1153 hash=52238f..df4608
INFO [08-31 09:33:00.319] Successfully sealed new block number=1154 sealhash=3007ac..490bec hash=348269..8849fb elapsed=1.099s
INFO [08-31 09:33:00.319]   block reached canonical chain number=1147 hash=119a1b..95d1f0
INFO [08-31 09:33:00.394] Commit new sealing work number=1155 sealhash=6268eb..c331a5 uncles=0 txs=0 gas=0 fees=0 elapsed=75.506ms
INFO [08-31 09:33:00.394] Commit new sealing work number=1155 sealhash=6268eb..c331a5 uncles=0 txs=0 gas=0 fees=0 elapsed=80.494ms
INFO [08-31 09:33:00.403]   mined potential block number=1154 hash=348269..8849fb
INFO [08-31 09:33:00.984] Successfully sealed new block number=1155 sealhash=6268eb..c331a5 hash=512bab..6a9534 elapsed=664.978ms
INFO [08-31 09:33:00.984]   block reached canonical chain number=1148 hash=5a36b6..049741
INFO [08-31 09:33:01.023] Commit new sealing work number=1156 sealhash=fcd4b4..b4248f uncles=0 txs=0 gas=0 fees=0 elapsed=39.423ms
INFO [08-31 09:33:01.027] Commit new sealing work number=1156 sealhash=fcd4b4..b4248f uncles=0 txs=0 gas=0 fees=0 elapsed=43.676ms
INFO [08-31 09:33:01.031]   mined potential block number=1155 hash=512bab..6a9534
INFO [08-31 09:33:01.881] Successfully sealed new block number=1156 sealhash=fcd4b4..b4248f hash=33517e..17b4ce elapsed=896.664ms
INFO [08-31 09:33:01.881]   block reached canonical chain number=1149 hash=bdee0d..47a547
INFO [08-31 09:33:01.887] Commit new sealing work number=1157 sealhash=598ae1..ba5988 uncles=0 txs=0 gas=0 fees=0 elapsed=5.703ms
INFO [08-31 09:33:01.890] Commit new sealing work number=1157 sealhash=598ae1..ba5988 uncles=0 txs=0 gas=0 fees=0 elapsed=8.777ms
INFO [08-31 09:33:01.893]   mined potential block number=1156 hash=33517e..17b4ce

```

```
C:\Administrator> C:\Windows\System32\cmd.exe - geth attach \\.pipe\geth.ipc  
(c) Microsoft Corporation. All rights reserved.  
  
C:\ProgramData\Microsoft\Windows\Start Menu\Programs>Geth>geth attach \\.pipe\geth.ipc  
Welcome to the Geth JavaScript console!  
  
instance: Geth/v1.10.25-stable-69568c55/windows-amd64/go1.18.5  
coinbase: 0x41551c89df8faec8fc8b5405130302686ecfdb81  
at block: 0 (Thu Jan 01 1970 05:30:00 GMT+0530 (IST))  
datadir: E:\MyNetwork\MyBlockchain  
modules: admin:1.0 debug:1.0 engine:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0  
  
To exit, press ctrl-d or type exit  
> eth.accounts  
[\"0x41551c89df8faec8fc8b5405130302686ecfdb81\"]  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
0  
> miner.start()  
null  
> eth.mining  
true  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
5100000000000000000  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
5500000000000000000  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
8750000000000000000  
>  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
1.585e+21  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
2.915e+21  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
2.93e+21  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
1.45e+21  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
4.67e+21  
> miner.stop()  
null  
> eth.getBalance('0x41551c89df8faec8fc8b5405130302686ecfdb81')  
5.095e+21  
> eth.mining  
false  
>
```

**Aim :** Demonstrate working of a blockchain. Create a blockchain in Python

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.getblockchaininfo</b><br>{<br>"chain": "main",<br>"blocks": 4991,<br>"headers": 964971,<br>"bestblockhash":<br>"00000000fa413253e1d30ff687f239<br>b528330c810e3de86e42a53817568<br>2599d",<br>"difficulty": 1,<br>"time": 1235132300,<br>"mediantime": 1235130482}<br>                                                                                                                                                                                                                                                                                                                              | <b>2.getmininginfo</b><br>{<br>"blocks": 5443,<br>"difficulty": 1,<br>"networkhashps":<br>5367449.178955262,<br>"pooledtx": 0,<br>"chain": "main",<br>"warnings": ""<br>}<br>                                                                                                                                                                                                                                                                                                                                                                                   | <b>3.getpeerinfo</b><br>[[<br>{"id": 0,<br>"addr": "169.155.169.220:8333",<br>"addrbind": "192.168.52.16:63246",<br>"addrlocal": "103.105.11.30:63246",<br>"network": "ipv4",<br>"services": "0000000000000c09",<br>"servicesnames": [<br>"NETWORK",<br>"WITNESS",<br>"NETWORK_LIMITED",<br>"UNKNOWN[2^11]"<br>]}<br>]]<br>                                                                                                                                                          |
| <b>4.getnetworkinfo</b><br>{<br>"version": 250000,<br>"subversion": "/Satoshi:25.0.0/",<br>"protocolversion": 70016,<br>"localservices": "0000000000000408",<br>"localservicesnames": [<br>"WITNESS",<br>"NETWORK_LIMITED"<br>],<br>"localrelay": true,<br>"timeoffset": 0,<br>"networkactive": true,<br>"connections": 10,<br>"connections_in": 0,<br>"connections_out": 10,<br>"networks": [{<br>"name": "ipv4",<br>"limited": false,<br>"reachable": true,<br>"proxy": "",<br>"proxy_randomize_credentials": false<br>}]<br>}<br><b>5.getbalance : 0.000000000</b><br><b>6.getblockcount : 108961</b> | <b>8. getblock</b><br>{<br>"hash":<br>"000000000000cc248cc894f96ff61ed10<br>97751902bd3c0b79bdc8d7c4cfe28b3",<br>"confirmations": 2661,<br>"height": 114834,<br>"version": 1,<br>"versionHex": "00000001",<br>"merkleroot":<br>"3b9af055f33bb5e47d822a88d6ab04ad<br>55ca8d267cab549ac3552615cb2b91b3"<br>"time": 1300971450,<br>"mediantime": 1300968921,<br>"nonce": 1542782224,<br>"bits": "1b00dc31",<br>"difficulty": 76192.61934751371,<br>"chainwork":<br>"00000000000000000000000000000000<br>000000000000000000024c517414385928<br>9",<br>"nTx": 3}<br> | <b>7. Getbestblockhash :</b><br>000000000000cc248cc894f96ff61ed109<br>7751902bd3c0b79bdc8d7c4cfe28b3<br><b>9. getblockheader</b><br>{<br>"hash":<br>"0000000000007d92fd811ca1fbe7c6<br>cfb6b65d3c6f32ff540357c6862aad6f<br>95",<br>"confirmations": 9168,<br>"height": 114833,<br>"version": 1,<br>"versionHex": "00000001",<br>"merkleroot":<br>"6d997e82147505cc883334444e53<br>482f451c20bbb082504a73c8ba0287<br>ae5c40",<br>"time": 1300971197,<br>"mediantime": 1300967599}<br> |
| <b>10. Getwalletinfo</b><br>{<br>"walletname": "TYCS2026",<br>"walletversion": 169900,<br>"format": "sqlite",<br>"balance": 0.00000000,<br>"unconfirmed_balance": 0.00000000,<br>"immature_balance": 0.00000000,<br>"txcount": 0,<br>"keypoolsize": 4000,<br>"keypoolsize_hd_internal": 4000,<br>"unlocked_until": 0,<br>"paytxfee": 0.00000000,<br>"private_keys_enabled": true,<br>"avoid_reuse": false,<br>"scanning": false,<br>"descriptors": true,<br>"external_signer": false<br>}<br>                                                                                                            | <b>11. getmempoolinfo</b><br>{<br>"loaded": true,<br>"size": 0,<br>"bytes": 0,<br>"usage": 0,<br>"total_fee": 0.00000000,<br>"maxmempool": 300000000,<br>"mempoolminfee": 0.00001000,<br>"minrelaytxfee": 0.00001000,<br>"incrementalrelayfee": 0.00001000,<br>"unbroadcastcount": 0,<br>"fullrbf": false<br>}<br>                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

